Alexander Metzger

Seattle, WA 98195

https://sandergi.com

alex@sandergi.com

My research aims to bridge gaps in information access for underserved languages and communities through a combination of algorithm design, machine learning, and edge device technology.

Education

UC Berkeley – *PhD in CS* 09/26 – 06/31

Advised by Professor Gopala. Systems and machine learning for accessible, resource-constrained, underserved-population computing with a focus on inclusive speech technology.

University of Washington – *3.99 GPA, Honors in Math* 09/22 – 06/26

Joint BS/MS Computer Science and BS Mathematics, Graduate Compilers & Cloud Security Teaching Assistant, UWSO First Violin, Accelerated Honors Math, Guest Lecturer [T8, T9, T10], NeurIPS+ICML+UIST reviewer.

Awards

- **[H4] CRA Outstanding Undergraduate Researcher Award**: Honorable Mention — 2025
- **[H3] NCWIT AiC Collegiate Award**: National Finalist (Top 63) — 2025
- **[H2] Departmental Honors and Dean's List**: Annual and Quarterly — 2022-2026
- **[H1] Akvelon Mentor's Choice**: Outstanding Leadership — 2023

Funding

- **[F7] National Science Foundation**: GRFP Fellow, $159,000 — 2026
- **[F6] Mozilla Builders Accelerator** – *$39K research grant, $35K cloud credits* — 2024

  Youngest out of 14 founders and researchers selected from 44 countries & 100s of applicants. Featured on the Stack Overflow podcast with more than 2 million viewers. Praised as "most promising" by Silicon & Pulse.
- **[F5] Microsoft Imagine Cup**: International Semifinalist, $5K cloud credits — 2025
- **[F4] Facebook AI for Health Challenge**: 1st Place, $1K Prize — 2025
- **[F3] Google Research**: Google TPU Research Cloud Access — 2025
- **[F2] Nvidia Inception**: Accepted Startup Founder, GPUs — 2025
- **[F1] Clean Energy Institute**: TARs Research Grant — 2025

Research Experience

Speech Lab – *Advisor: Dr. Gopala Krishna Anumanchipalli* Jun. 2026 – Present

Transfer Learning from L2 Intervention Data for Dysarthric and Impaired Speech. Clinical Speech.

Goal: Investigate whether models can generalize from L2 to speech pathology settings.

- **Collected** a large dataset with >2K unique speakers across 139 native language backgrounds and 1M pronunciation intervention trajectories.
- **Coordinating** collaboration between UC Berkeley, Koel Labs, and UT Austin.
- **Leading** grant proposals and experiment design.

Ubiquitous Computing Lab – *Advisors: Dr. Shwetak Patel & Dr. Vikram Iyer* Jun. 2024 – Aug. 2026

Embedded Arena: Iterative Optimization via Hardware Feedback. Embedded ML. Link.

Goal: An Embedded ML benchmark that is hard to saturate and involves hardware-in-the-loop.

- **Designed** a novel benchmark for agent ability to respond to multiple-turn physical-hardware feedback.
- **Co-first-authored** a paper [P4] proposing the benchmark and a new hardware-in-the-loop harness that brings agent success rate from 0% to up to 80% for hard resource-contrained model deployment tasks. Examined applications: animal monitoring and wearables to assess linguistic development in toddlers.
- **Created** SoTA speech2ipa models that retain top performance with 600x smaller memory footprint.

Multi-modal automated lifecycle assessments of consumer electronics. ML for sustainability. Link.

Goal: Democratize access to environmental impact estimates and empower sustainable consumption.

- **Co-authored** study design and research papers [P3, P5] and presented the work at top events [T11, T14].
- **Industry adoption** of our work by Google and Amazon as well as a publicly accessible Chrome Extension.
- **Designed** the multi-modal pipeline, enabling agents to consolidate information from many fragmented sources and departments (online tear downs, FCC records, environmental reports, company data).

ICTD Lab – *Advisor: Dr. Richard Anderson* Sep. 2022 – May 2024

eKichabi v2: Digital phone directory for subsistence farmers in rural Tanzania. ICTD. Link.

Goal: Study the economic impacts of digital agricultural information systems in Sub-Saharan Africa.

- **Co-first-authored** the paper [P6] accepted to ACM CHI 2024. Presented talks [T8, T13].
- **Led** team of 4 developers for the USSD server and later also the Android App optimizing latency by 40%.
- **Designed** a binary protocol reducing data costs by 70%, making the platform accessible to 10K farmers.

DGP Group – *Advisor: Dr. Farhan Samir* Sep. 2024 – Present

Transcribing in Context: Temporal Trends in ASR Biases. Inclusive Speech Technology. Link.

Goal: Re-evaluate the claim that speech models are becomming more universal in context of dialects.

- **First-authoring** the paper [P7], mentoring an undergrad and a high school student.
- **First** comprehensive evaluation of ASR model regressions over time for minority speaker profiles.
- **Assembled** a diverse research team from Korea, England, Denmark, Japan, India, Canada, and the US.
- **Quantifying** real world impact on immigration processes, job acquisition, and information access.

Towards Universal Phoneme Recognition. Computational Linguistics. Link.

Goal: Device an inclusive benchmark for phoneme recognition and design a universal model.

- **Established** the current standard phoneme recognition leaderboard and trained the top model.
- **Mentoring** high school and undergraduate students and establishing research collaborations.
- **First-authored** a paper [P8] establishing a large benchmark for accented and impaired English speech and systematically evaluating data quality and quantity scaling trade-offs.

University of Washington – *Advisor: Dr. Stefan Steinerberger* Sep. 2024 – Present

Practical algorithms for graph embedding. Graph Theory, Algorithms, Combinatorics. Link.

Goal: Compute the genus of the previously intractable (3, 12)-cage graph (turns out it is 17).

- **Invented and implemented** SoTA algorithm in C with visualization and verification in Python.
- **Started as independent research** and then reached out to Professor Steinerberger and Brinkmann.
- **Co-first authored** paper submitted to Discrete Mathematics, published thesis, and gave talks [T4, T7].

Publications

[P8] – *Interspeech* Phonetic Transcription, Speech ML

Alexander Metzger, Aruna Srivastava, Ruslan Mukhamedvaleev. 2026. Scaling Human and G2P Supervision for Robust Phonetic Transcription.

[P7] – *Pre-print* Automatic Speech Recognition, Fairness and Equity

Alexander Metzger*, Aruna Srivastava*, Ruslan Mukhamedvaleev, Eunjung Yeo, Ishtiaque Ahmed, Sachin Kumar, Nina A. Markl, Farhan Samir. Connecting the Dots: Enduring Performance Gaps in ASR Systems Between Standard and Minority English Varieties.

[P6] – *ACM CHI* HCI, ICTD

Ananditha Raghunath*, **Alexander Metzger***, Hans Easton, XunMei Liu, Fanchong Wang, Yunqi Wang, Yunwei Zhao, Hosea Mpogole, and Richard Anderson. 2024. eKichabi v2: Designing and Scaling a Dual-Platform Agricultural Technology in Rural Tanzania.

[P5] – *Nature Electronics* ML for Sustainability, CV, Agents

Zhihan Zhang*, **Alexander Metzger***, Yuxuan Mei, Felix Hähnlein, Zachary Englhardt, Tingyu Cheng, Gregory D. Abowd, Shwetak Patel, Adriana Schulz, Vikram Iyer. 2026. Sustainability assessment using multimodal artificial intelligence agents.

[P4] – *Pre-print* Embedded ML, Benchmark, Agents

Zhihan Zhang*, **Alexander Metzger***, Jiuyang Lyu*, Chun-Cheng Chang, Jiayi Shao, Yujia Liu, Emmanuel Azuh Mensah, Edward Wang, Kurtis Heimerl, Gregory D. Abowd, Shwetak Patel, Natasha Jaques, Vikram Iyer. Embedded Arena: Iterative Optimization via Hardware Feedback.

[P3] – *ACM IMWUT* ML for Sustainability, CV, HCI

Zhihan Zhang, Puvarin Thavikulwat, **Alexander Metzger**, Yuxuan Mei, Felix Hähnlein, Zachary Englhardt, Gregory D. Abowd, Shwetak Patel, Adriana Schulz, Tingyu Cheng, and Vikram Iyer. 2025. Living Sustainability: In-Context Interactive Environmental Impact Communication.

[P2] – *Pre-print, submitted to Discrete Mathematics* Algorithms, Graph Combinatorics

Alexander Metzger* and Austin Ulrigg*. An Efficient Genus Algorithm Based on Graph Rotations.

[P1] – *University of Washington Thesis* Topology, Graph Theory, Algorithms

Alexander Metzger. A Practical Algorithmic Approach to Graph Embedding. University of Washington. Department of Mathematics. Honors Thesis. https://hdl.handle.net/1773/53829.

TALKS

[T14] – *In-Context Interactive Environmental Impact Communication* Oct. 2025

DUB Research Day, Seattle, WA [100s of attendees, only undergrad of 8 invited speakers]

[T13] – *eKichabi v2: Designing and Scaling a Dual Platform Technology in Rural Tanzania* May 2024

DUB Para.chi Event, Seattle, WA [100s of attendees, youngest invited speaker]

[T12] – *Deploying Speech Technology at Scale* Aug. 2025

Interspeech Conference, Rotterdam, Netherlands [2K attendees, 1 of 4 speakers invited for the science slam]

[T11] – *In-Context Interactive Environmental Impact Communication* Oct. 2025

Paul G. Allen's Annual Research Showcase, Seattle, WA [100s of attendees, 3 posters and a talk]

[T10] – *Embedded ML - Computer Vision Demo* Oct. 2025

University of Washington Embedded Systems Capstone Guest Talk, Seattle, WA [20 students]

[T9] – *Computational Linguistics for Machine Aided Pronunciation Learning* Oct. 2024

University of Washington Computational Linguistics Group Guest Talk, Seattle, WA [30 PhD attendees]

[T8] – *Designing and Deploying Digital Information Systems in Sub-Saharan Africa* Oct. 2023

University of Washington CHANGE Seminar Guest Talk, Seattle, WA [40 PhD attendees]

[T7] – *Graph Algorithms and Optimization* Nov. 2025

Northwest Undergraduate Mathematics Symposium, Bothell, WA [30 attendees]

[T6] – *On-device ML for Accessible and Privacy-Preserving Concussion Care* May. 2026

University of Washington Research Symposium, Seattle, WA [60 attendees]

[T4&T5] – *Graph Embedding and Genus* | *Speech Technology Built For Everyone, Everwhere* May. 2025

University of Washington Research Symposium, Seattle, WA [60 attendees]

[T2&T3] – *The Future of Language Learning* Sep. & Dec. 2024

Mozilla Builders, New York, NY | Mozilla Builders, San Francisco, CA [500 attendees]

[T1] – *Building Inclusive Speech Technology* Apr. 2025

DubHacks Next Demo Day, Seattle, WA [100 attendees]

Work Experience

Koel Labs – *Founder and CEO* Aug. 2024 – Present

- **Raised $100K in funding** (Mozilla, Microsoft, Google, Nvidia) for computational linguistics research.
- **Trained** SoTA NLP models, cutting streaming latency by 60% and doubling accuracy [P8].
- **Managed** open-source team of 4 engineers and 6 researchers from CMU, UToronto, UT Austin, and UBC.
- **Established** strategic partnerships and research collaborations with 5 companies and 5 institutions.

Gooey.AI – *Software Engineer* Jun. 2023 – Sep. 2024

- **Led** client communication and 4-member intern team, merging 70+ PRs across production ML pipelines.
- **Deployed** ML solutions to 10M+ farmers across 5+ countries, demoed to 193 world leaders at the UN.

Akvelon, Inc. – *Software Development Intern* Jun. 2023 – Aug. 2023

- **Led** team of 4 to make 200K open-source contributions to 7 Microsoft repos across 27 pull requests.

Leadership

Seattle Tutoring Partners – *Founder and Research Mentor* May. 2022 – Aug. 2026

- **Mentoring** research projects for High Schoolers, and collaborating on internationaly recognized projects.
- **Developed** software for 500+ youth musicians and antenna software for 10K+ aspiring scientists.

Cascade Enrichment – *Technical Lead* Sep. 2022 – Sep. 2024

- **Managing** certification and curriculum for 30+ tutors; developed online platform for 100+ students.

Design Build Fly – *Lead Computer Scientist* Sep. 2023 – Sep. 2024

- **Coordinated** optimization algorithms and data analysis for 30 engineers; placing 3rd nationally.

Relevant Coursework

HCI: Capstone Software Design to Empower Underserved Populations, Software Design and Implementation, Social Networks, Photography, Robotics Colloquium, Computer Science Colloquium

Systems: Compilers, Operating Systems, Computer Communication Networks, Distributed Systems, Systems Programming, Hardware/Software Interface, Computer Systems Architecture, Computer Graphics, Concurrency/Parrallelism and Rust, Databases, Digital Design, Coursera Nand To Tetris

ML: Capstone Natural Language Processing, Computational Biology, Probabilistic Robotics, Probability and Statistics, Reinforcement Learning, Computer Vision, Machine Learning by Andrew Ng

Theory: Theory of Computation, Algorithms, Modern Algorithms, Data Structures and Parallelism, Numerical Analysis, Combinatorics, Quantum Information/Computation, Cryptography, Modern Algebra and Coding Theory, Linear Algebra, Differential Equations, Vector Calculus